UNIT 2 · 9 hours

EMF Equation of a Transformer



Transformers and Electrical Machines

Basic Electrical and Electronics Engineering (EEE103)
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Learning objectives

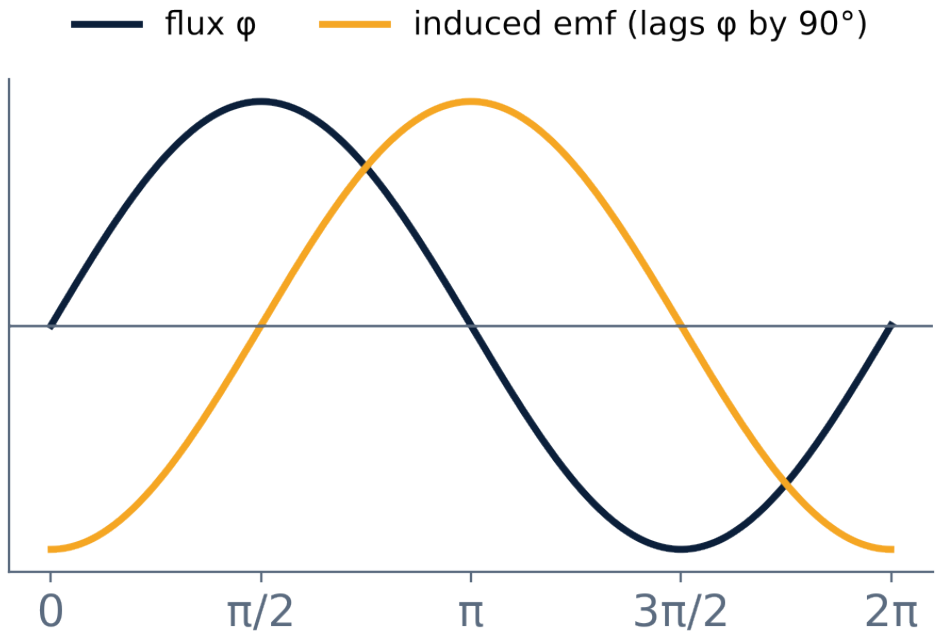
- Derive the transformer EMF equation
- Calculate turns and maximum flux
- Find flux density from core area
- Solve numericals on transformer EMF and ratio

Syllabus context

Transformers: necessity of transformers, principle of operation, types and construction of single-phase transformers, EMF equation, losses and efficiency.

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Flux and induced EMF



- Primary flux φ = φm sin ωt.
- e = −N dφ/dt = −N ω φm cos ωt.
- EMF lags the flux by 90°.
- Peak EMF Em = 2π f N φm.

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Deriving the EMF equation

1

Flux change

Flux rises from 0 to φm in a quarter cycle and from +φm to −φm in 1/2f s.

2

Average rate

2φm ÷ (1/2f) = 4 f φm weber per second.

3

EMF per turn

Average EMF per turn = 4 f φm volts.

4

RMS

Multiply by form factor 1.11: 4.44 f φm per turn.

E1 = 4.44 f φm N1 and E2 = 4.44 f φm N2 (RMS volts)

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Using the EMF equation

$$E_1 = 4.44\, f\, \varphi_m\, N_1$$

primary

$$E_2 = 4.44\, f\, \varphi_m\, N_2$$

secondary

$$E_1/E_2 = N_1/N_2$$

ratio (flux and f cancel)

$$B_m = \varphi_m / A$$

flux density (T) for core area A

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Numerical 1: 200 kVA transformer

Problem 200 kVA, 3300/230 V, 50 Hz single-phase transformer has 80 turns on the secondary. Find the approximate primary and secondary currents, primary turns and maximum flux.

- $I_1 = 200 \times 1000/3300 = 60.6$ A; $I_2 = 200 \times 1000/230 = 869.6$ A
- $N_1 = N_2 \times (V_1/V_2) = 80 \times 3300/230 = 1148$ turns
- $\varphi_m = E_2/(4.44\, f\, N_2) = 230/(4.44 \times 50 \times 80)$

Answer: $I_1 = 60.6$ A, $I_2 = 869.6$ A, $N_1 = 1148$, $\varphi_m = 12.9$ mWb

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Numerical 2: flux density

Problem A 50 Hz transformer has 80 primary and 400 secondary turns, net core area 200 cm^2. Primary is on 230 V. Find E2 and Bm.

- $K = 400/80 = 5;\ \ E2 = 5 \times 230 = 1150$ V
- $\varphi m = 1150/(4.44 \times 400 \times 50) = 12.95$ mWb
- $Bm = \varphi m/A = 12.95\times10^{-3} / (200\times10^{-4})$

Answer: E2 = 1150 V, Bm ≈ 0.65 T

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Numerical 3: turns required

Problem A 25 kVA, 50 Hz, 2000/200 V transformer has φm = 0.045 Wb. Find the turns on each winding.

- $N_2 = E_2/(4.44\, f\, \varphi_m) = 200/(4.44 \times 50 \times 0.045) = 20.02 \approx 20$
- $N_1 = N_2 \times E_1/E_2 = 20 \times 2000/200$

Answer: N2 = 20 turns, N1 = 200 turns

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Key takeaways

- E = 4.44 f φm N (RMS)
- Average emf per turn = 4 f φm; multiply by 1.11 for RMS
- E1/E2 = N1/N2
- Bm = φm / core area

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Practice questions

Q1. Derive E = 4.44 f φm N.

Q2. A 1100/220 V, 50 Hz transformer has 55 secondary turns. Find N1 and φm.

Q3. Core area 150 cm², Bm = 1.2 T, 50 Hz, 500 turns: find E.

Q4. Why does the same core flux serve both windings?